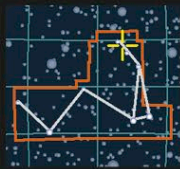


DOUBLE STAR

15 Monocerotis



DISTANCE FROM SUN
1,020 light-years

MAGNITUDE 4.7

SPECTRAL TYPE O7

MONOCEROS

15 Monocerotis (15 Mon), also known as S Monocerotis, is an O-type binary system located within the open cluster NGC 2264. It is a blue supergiant star—young, massive, and about 8,500 times more luminous than the Sun. It is also a variable star, exhibiting a small (0.4 magnitude) change in brightness. 15 Mon is responsible for illuminating the Cone Nebula (see p.242) and consequently is an easy target for amateur astronomers. 15 Mon is an astrometric and spectroscopic binary—that is, its companion star is detected through observations of the motion of 15 Mon, and also through spectroscopic analysis of 15 Mon's starlight (see p.274). The companion orbits 15 Mon with a period of 24 years, and recent studies using the Hubble Space Telescope show that the closest approach between the stars occurred in 1996. It has been suggested that 15 Mon is a multiple system, with three other bright giants nearby, but there is no evidence that the other giants are associated with 15 Mon.



BRIGHTEST STAR

The brightest star in the open star cluster NGC 2264, 15 Monocerotis sits in close visual proximity to the Cone Nebula (see p.242).

BRILLIANT ILLUMINATION

Even the most powerful telescopes cannot separate the two stars of 15 Monocerotis visually. The brilliant blue star lights up the emission nebula that surrounds it.